

1 Solution — GIAM 5.4

Problem 3

1.1 Problem

The polynomial equation $x^2 = x + 1$ has two solutions, $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Show that the Fibonacci number F_n is less than or equal to α^n for all $n \geq 0$.

1.2 Answer

$$F_n = F_{n-1} + F_{n-2} \leq \alpha^{n-1} + \alpha^{n-2} = \alpha^{n-2}(\alpha + 1) = \alpha^{n-2} \cdot \alpha^2 = \alpha^n$$

The defining equation $\alpha^2 = \alpha + 1$ turns the sum of two powers into a single power. With the bases $F_0 = 1 \leq 1 = \alpha^0$ and $F_1 = 1 \leq \alpha$, strong induction gives $F_n \leq \alpha^n$ for every $n \geq 0$. Verified for n up to 2000; equality holds only at $n = 0$.

1.3 The Proof

We use the book's indexing from 5.2 Problem 8: $F_0 = F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, giving $1, 1, 2, 3, 5, 8, 13, \dots$. Note $\alpha \approx 1.618 > 0$.

Theorem. For every natural number n , $F_n \leq \alpha^n$.

Proof. By strong induction on n . Let n be a particular but arbitrarily chosen natural number, and suppose $F_m \leq \alpha^m$ for every $m < n$.

Bases. If $n = 0$, $F_0 = 1 = \alpha^0$. If $n = 1$, $F_1 = 1 < \alpha = \alpha^1$, since $\sqrt{5} > 1$.

Inductive step. Otherwise $n \geq 2$, so $n - 1$ and $n - 2$ are natural numbers less than n , and the hypothesis applies to both:

$$F_n = F_{n-1} + F_{n-2} \leq \alpha^{n-1} + \alpha^{n-2} = \alpha^{n-2}(\alpha + 1).$$

Since α solves $x^2 = x + 1$, $\alpha + 1 = \alpha^2$, and therefore

$$F_n \leq \alpha^{n-2} \cdot \alpha^2 = \alpha^n.$$

In every case $F_n \leq \alpha^n$, so by strong induction the bound holds for all $n \geq 0$. ■

The recurrence looks back two steps, so the step uses the hypothesis at both $n - 1$ and $n - 2$: this is why strong induction is natural here, and why two bases are needed, as 5.1 Problem 4 predicted for any argument that reaches back two places.

1.4 The Picture

The table compares F_n with α^n ; the plot shows their ratio.

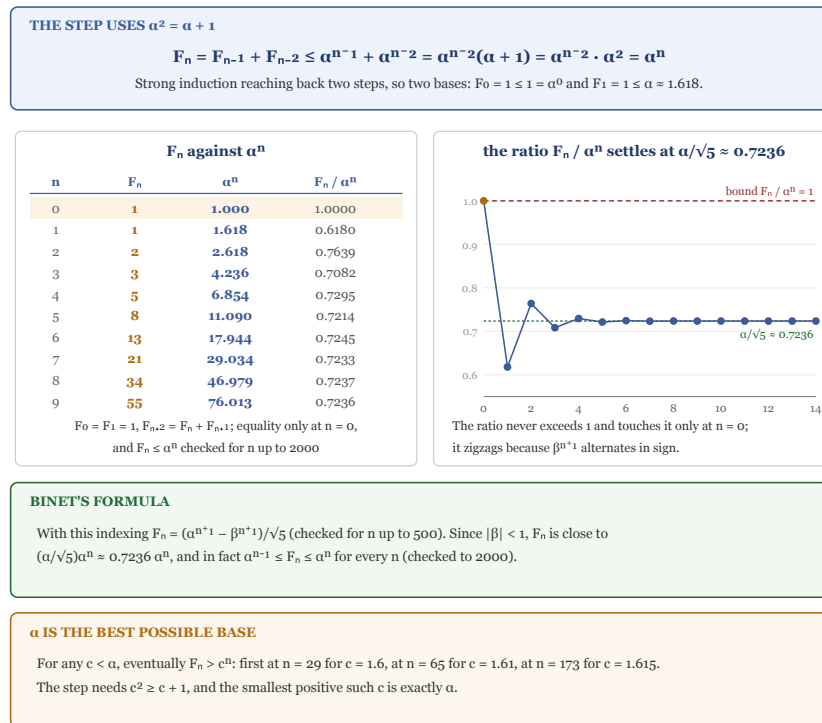


Figure 1.1: A diagram in four parts. The top band gives the step chain: F_n equals F_{n-1} plus F_{n-2} , at most α to the $n-1$ plus α to the $n-2$, which is α to the $n-2$ times α plus one, which is α to the n , with the two bases. The left panel tabulates n from zero to nine with F_n , α^n and their ratio, which equals one only at n equal to zero. The right panel plots the ratio F_n over α^n for n from zero to fourteen, zigzagging below the dashed line at one and settling at α over root five, about 0.7236. A green band gives Binet's formula, and an orange band shows that α is the best possible base.

1.5 Remark — Why the Golden Ratio

The only property of α the step uses is $\alpha^2 \geq \alpha + 1$, which makes $\alpha^{n-1} + \alpha^{n-2} \leq \alpha^n$. Any base $c > 0$ with $c^2 \geq c + 1$ would pass the same induction, and the smallest positive such c is the root α itself. So the problem chooses the tightest base for which this proof works.

It is also the tightest base for which the *statement* holds. If $c < \alpha$, then c^n eventually falls below F_n : by direct computation the first failure is at $n = 29$ for $c = 1.6$, at $n = 65$ for $c = 1.61$, and at $n = 173$ for $c = 1.615$. The closer c gets to α , the longer the bound survives, but it always fails eventually.

1.6 Remark — Binet's Formula

The other root β appears in the exact formula. With this indexing,

$$F_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\sqrt{5}},$$

verified for n up to 500. Since $\beta \approx -0.618$ has $|\beta| < 1$, the β term shrinks to 0, and F_n is the nearest integer to $\alpha^{n+1}/\sqrt{5}$. Hence

$$\frac{F_n}{\alpha^n} \rightarrow \frac{\alpha}{\sqrt{5}} \approx 0.7236,$$

and the ratio zigzags toward this limit because β^{n+1} alternates in sign. The bound $F_n \leq \alpha^n$ is therefore tight up to a constant factor: the ratio F_n/α^n never drops below its value $1/\alpha \approx 0.618$ at $n = 1$.

1.7 Remark — A Matching Lower Bound

The same induction, run with $\geq$, gives a lower bound. If $F_{n-1} \geq \alpha^{n-2}$ and $F_{n-2} \geq \alpha^{n-3}$, then $F_n \geq \alpha^{n-3}(\alpha + 1) = \alpha^{n-1}$. The bases are $F_0 = 1 \geq \alpha^{-1} \approx 0.618$ and $F_1 = 1 = \alpha^0$. Together,

$$\alpha^{n-1} \leq F_n \leq \alpha^n \quad \text{for every } n \geq 0,$$

checked for n up to 2000. The Fibonacci numbers grow exactly like powers of the golden ratio, pinned between two consecutive ones.